

6. Amorphous polymers

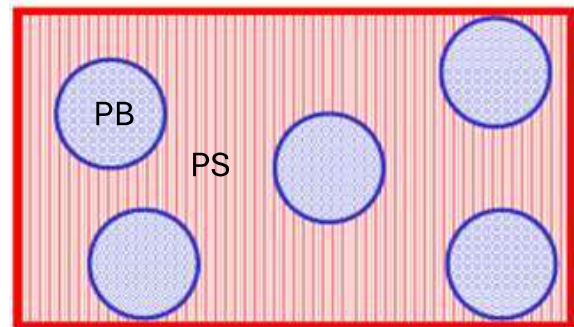
6.2 Polystyrene and styrenics

High-impact polystyrene (HIPS)

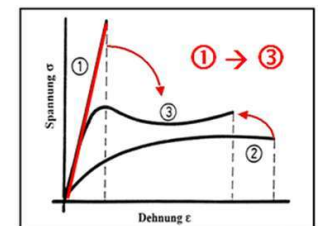
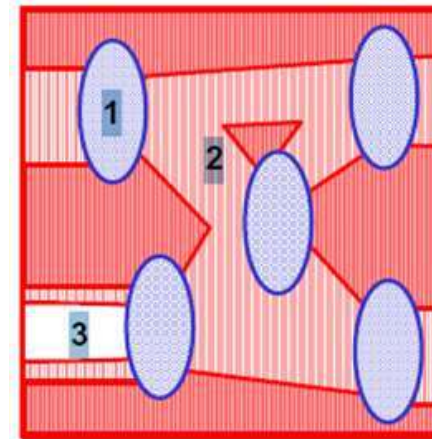
The principle of toughness enhancement (toughening mechanisms)

Micromechanical mechanisms of toughness enhancement by means of rubber particles

1. **initiation of multiple crazing** by stress concentration in the glassy matrix (PS)
2. **crack stop** (stop of crack propagation) at rubber particles
3. **crack bridging** and stabilization by rubber particles



PB rubber particles dispersed in the PS matrix



1 ... mechanical characteristics of PS
2 ... mechanical characteristics of PB
3 ... the aim of toughening